

Mayo Clinic Study of Aging

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1 Initial look

- Population: the set of subjects that you would like to write about
- Frame: the set who have an opportunity to be sampled
- Sample: the study data set
- $Y(t)$ = under observation at time t
- MCSA
 - Frame: All residents of Olmsted County, Minnesota. A complete census is available through the Rochester Epidemiology Project
 - Sample: An age and sex stratified sample was invited to join the study, those who agree were scheduled for an intake visit. Return visits at 15 month intervals.
 - At each visit, a subject is invited to participate in imaging. About 40% agree.
 - See Roberts et al (PMID:18259084) for details.

```
> library(mshmm)
> table2 <- function(...) table(..., useNA= "ifany")
>
> # dim(mcsa)
> # 40939 294
> dim(dementia)
[1] 25671    12
> table(table(dementia$ptnum))

  2   3   4   5   6   7   8   9  10  11  12  13  14  15
429 613 538 462 496 581 469 268 125 104 123  92  16   1
> subset(dementia, ptnum<3)
  ptnum  age male educ any.e4 cmc  dx centiloid p.tau181  hpvol
1     1  69.2   0  13     0   1  CU         NA         NA     NA
2     1  70.1   0  13     0   1  CU         NA         NA     NA
```

```

3      1 71.2    0  13      0  1    CU      NA      NA      NA
4      1 72.9    0  13      0  1    CU      NA      NA      NA
5      1 76.0    0  13      0  1    CU      NA      NA      NA
6      1 76.7    0  13      0 NA alive    NA      NA      NA
7      2 59.9    1  20      0  1    CU      4      NA 8.686971
8      2 61.0    1  20      0  1    CU      NA      NA      NA
9      2 62.5    1  20      0  1    CU      2      9.122 8.843209
10     2 63.9    1  20      0  1    CU      11     NA 8.629439
11     2 64.8    1  20      0  1    CU      NA      NA      NA
12     2 65.9    1  20      0  1    CU      3     15.885 8.268467
13     2 67.5    1  20      0  1    CU      NA      NA      NA
14     2 67.8    1  20      0 NA alive    NA      NA      NA

```

```

      memory      global
1  0.2876302  0.144226854
2  0.2134226 -0.001177957
3 -0.1168855 -0.106892374
4  0.3111153  0.400552893
5  0.1585472  0.137021188
6      NA      NA
7  1.2454318  1.580173450
8      NA  2.273353239
9  1.8710097  2.059384119
10 1.8808997  1.926561937
11      NA  1.999232218
12 2.1123333      NA
13 2.4916771  1.854048653
14      NA      NA

```

```
> table2(dementia$dx)
```

```

      alive      CU      MCI dementia      nondem      off-dem      death
      2680     17264     2357      395      1074      430      1471

```

```
> first <- subset(dementia, !duplicated(ptnum))
```

```
> table2(first$dx)
```

```

      alive      CU      MCI dementia      nondem      off-dem      death
      0      3785     532      0      0      0      0

```

```
> survcheck(Surv2(age, dx) ~1, id=ptnum, data=dementia)
```

```
Call:
```

```
survcheck(formula = Surv2(age, dx) ~ 1, data = dementia, id = ptnum)
```

```

Unique identifiers      Observations      Transitions
      4317      21354      4528

```

```
Transitions table:
```

from	to	CU	MCI	dementia	nondem	off-dem	death	(censored)
CU		0	845	30	815	184	11	2347
MCI		449	0	202	257	246	19	228
dementia		0	29	0	0	0	172	30
nondem		0	0	0	0	0	941	81
off-dem		0	0	0	0	0	328	102
death		0	0	0	0	0	0	0

Number of subjects with 0, 1, ... transitions to each state:

	count	0	1	2	3	4	5	6	7	9
state		0	1	2	3	4	5	6	7	9
CU		3912	363	40	2	0	0	0	0	0
MCI		3560	649	101	5	2	0	0	0	0
dementia		4094	214	9	0	0	0	0	0	0
nondem		3245	1072	0	0	0	0	0	0	0
off-dem		3887	430	0	0	0	0	0	0	0
death		2846	1471	0	0	0	0	0	0	0
(any)		2296	334	1125	380	123	47	9	2	1

```
> prog <- survfit(Surv2(age, dx) ~ male, dementia, id=ptnum,
  start.time= 73, p0=c(1,0,0,0,0,0))
```

```
> prog
```

```
Call: survfit(formula = Surv2(age, dx) ~ male, data = dementia, id = ptnum,
  start.time = 73, p0 = c(1, 0, 0, 0, 0, 0))
```

	n	nevent	rmean	se(rmean)*
male=0, CU	7820	168	14.2011304	0.26426170
male=1, CU	8223	213	12.9498004	0.24603512
male=0, MCI	7820	397	1.7093132	0.09837377
male=1, MCI	8223	416	1.6142819	0.09382098
male=0, dementia	7820	108	0.3884077	0.04660782
male=1, dementia	8223	123	0.3640890	0.04461705
male=0, nondem	7820	451	0.3836311	0.04215665
male=1, nondem	8223	553	0.2263417	0.02334722
male=0, off-dem	7820	225	0.7485480	0.06736584
male=1, off-dem	8223	201	0.4282215	0.04199303
male=0, death	7820	660	13.6689696	0.25973846
male=1, death	8223	780	15.5172655	0.24480653

*restricted mean time in state (max time = 104.1)

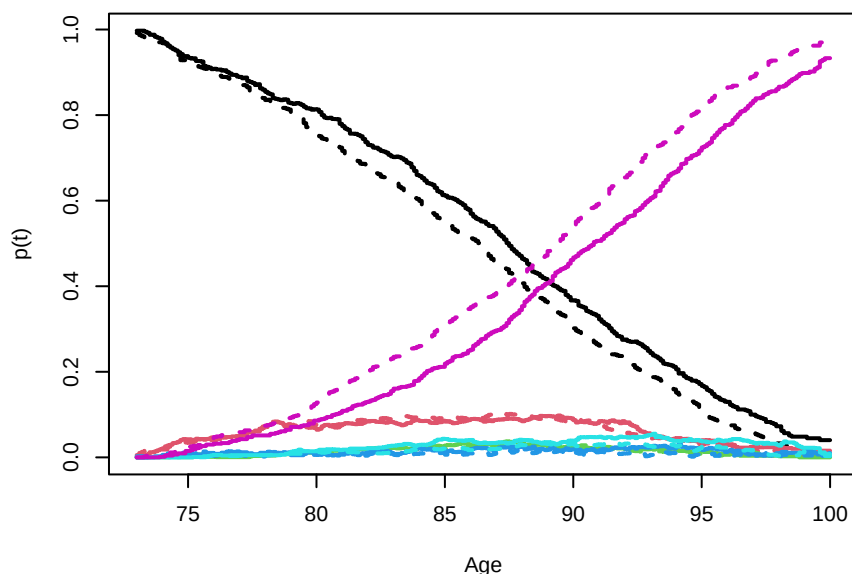
```
> dim(prog)
```

```
strata states
```

```
2 6
```

```
>
```

```
> plot(prog, lty=1:2, col=rep(1:6, each=2), lwd=2, xmax=100,
  xlab="Age", ylab="p(t)")
```



This isn't really what I want, which is curves of the underlying patient state, not curves of my diagnosis pathway. Start simpler with states of not demented (ND), dementia, and death.

```
> dem2 <- dementia
> temp <- c(1,2,2,3,2,3,4)[as.numeric(dem2$dx)]
> dem2$dstate <- factor(temp, 1:4, c("alive", "ND", "dementia", "death"))
> with(dem2, table2(dstate, dx))
```

dstate	alive	CU	MCI	dementia	nondem	off-dem	death
alive	2680	0	0	0	0	0	0
ND	0	17264	2357	0	1074	0	0
dementia	0	0	0	395	0	430	0
death	0	0	0	0	0	0	1471

```
> survcheck(Surv2(age, dstate, repeated=TRUE) ~1, dem2, id=ptnum)
```

Call:

```
survcheck(formula = Surv2(age, dstate, repeated = TRUE) ~ 1,
  data = dem2, id = ptnum)
```

Unique identifiers	Observations	Transitions
4317	21354	18674

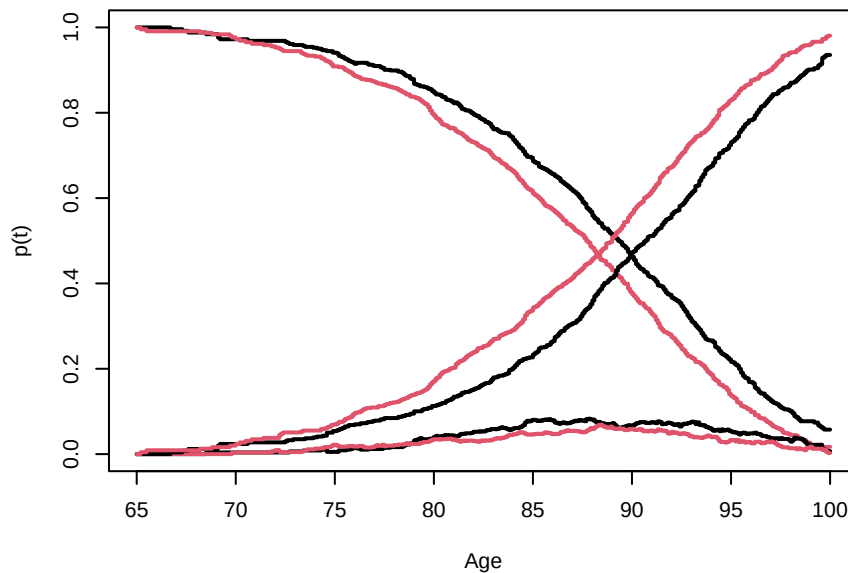
Transitions table:

from	to	ND	dementia	death	(censored)
ND		16349	662	971	2535
dementia		29	163	500	132
death		0	0	0	0

Number of subjects with 0, 1, ... transitions to each state:

	count	0	1	2	3	4	5	6	7	8	9	10	11	12	13
state		0	1	2	3	4	5	6	7	8	9	10	11	12	13
ND		496	672	531	448	475	543	478	229	130	99	108	93	15	0
dementia		3664	554	53	27	13	4	2	0	0	0	0	0	0	0
death		2846	1471	0	0	0	0	0	0	0	0	0	0	0	0
(any)		390	375	573	459	515	592	550	285	182	134	122	109	23	8

```
>
> dsurv <- survfit(Surv2(age, dstate) ~ male, dem2, id=ptnum,
+                  p0=c(1,0,0), start.time=65)
> plot(dsurv, col=1:2, lty=1, lwd=2, xmax=100,
+      xlab="Age", ylab="p(t)")
```



What is the effect of repeated=TRUE?

- survcheck: more information in the transitions table, count table is not useful
- survfit: none
- coxph: large. We are treating repeated ND visits like repeated infections

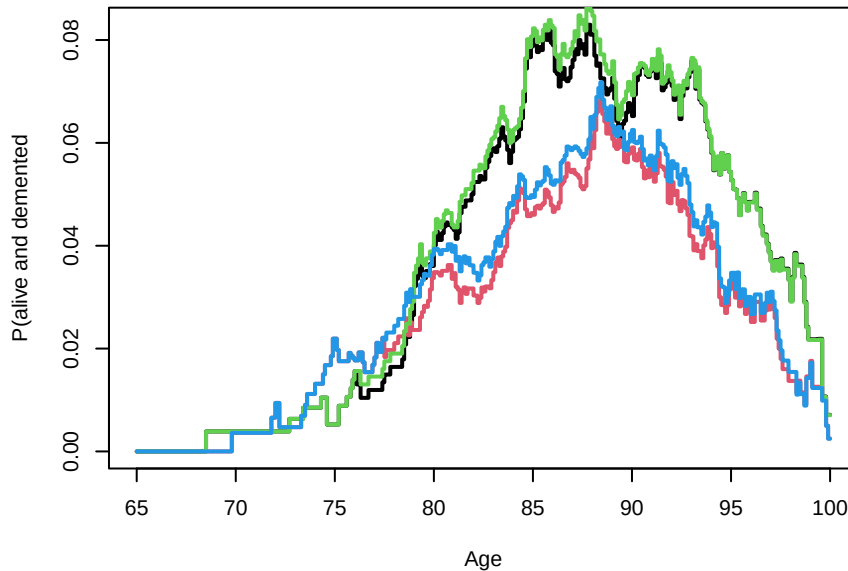
I'd like to recreate the 2022 analysis in Jack, Therneau, et al, PMID:35310829, and for that need to go one step further and use ever dementia (but not dead) as the second state. That is, no reversion from dementia to ND.

```
> # There must be a less mysterious way to create dstate2...
> temp1 <- as.numeric(dem2$dstate)
> temp2 <- unlist(tapply(temp1, dem2$ptnum,
```

```

function(x) ifelse(x==1, 1, cummax(x)))
> dem2$dstate2 <- factor(temp2, 1:4, c("alive", "ND", "dementia", "death"))
> survcheck(Surv2(age, dstate2) ~1, dem2, id=ptnum)$tran
      to
from    dementia death (censored)
  ND      653   957      2707
  dementia    0   514      138
  death       0    0         0
>
> dsurv2 <- survfit(Surv2(age, dstate2) ~ male, dem2, id=ptnum,
                    p0=c(1,0,0), start.time=65)
> plot(dsurv[,2], col=1:2, lwd=2, xmax=100,
       xlab="Age", ylab="P(alive and demented)")
> lines(dsurv2[,2], col=3:4, lwd=2)

```

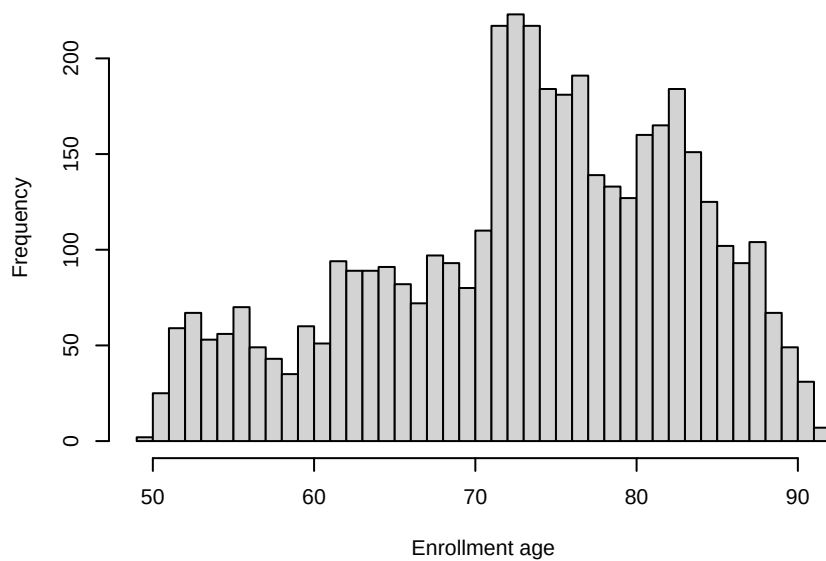


2 Age scale

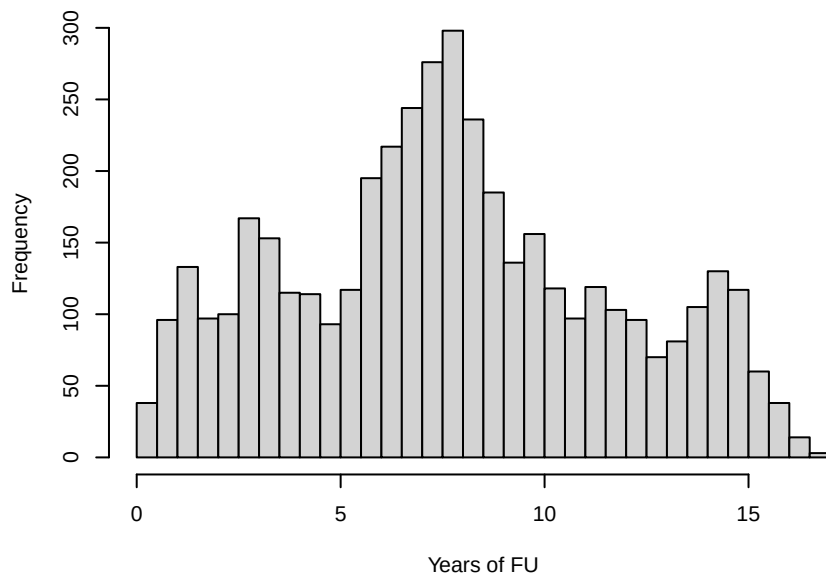
In epidemiology studies there is a common dichotomy of match vs model

- Match: Critical variable, we want to adjust for it in the most complete way. A coefficient or p-value is not needed
- Model: Describe the effect of this variable on the outcome via parameters.
- Common: match (stratify) on age and sex

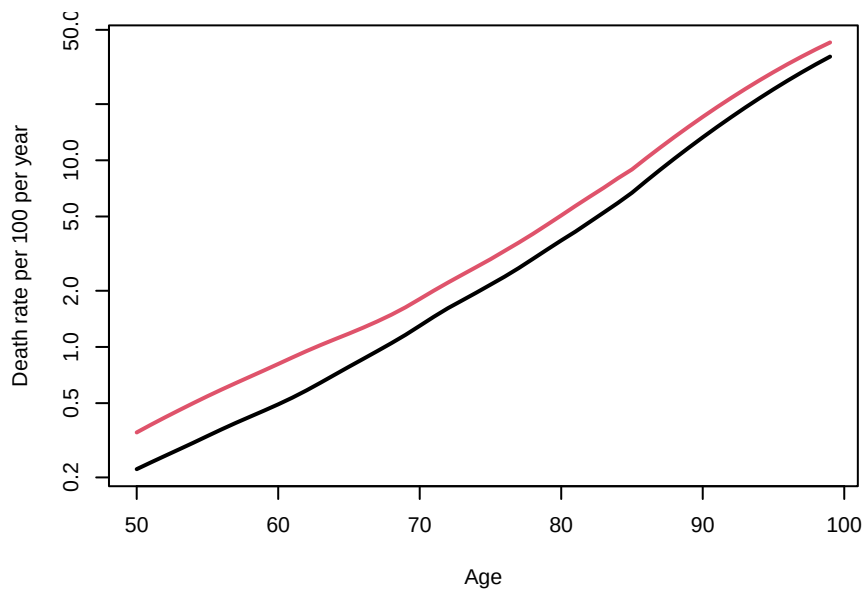
```
> first <- subset(dementia, !duplicated(ptnum))
> hist(first$age, nclass=30, xlab="Enrollment age", main=NULL)
```



```
> last <- subset(dementia, !duplicated(ptnum, fromLast=TRUE))
> hist(last$age-first$age, nclass=30, xlab="Years of FU", main=NULL)
```

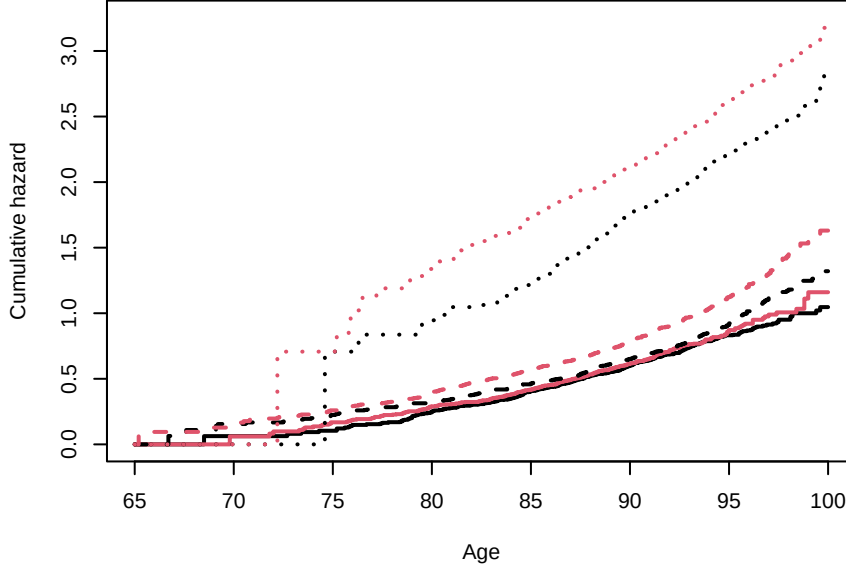


```
>
> matplot(50:99, 36525*survexp.mn[51:100,, "2015"], type='l', lty=1, lwd=2,
  col=2:1, log='y',
  xlab="Age", ylab="Death rate per 100 per year")
```



- Age is a dominating factor, we need to get it right, we don't need a p-value.
- For a coxph model on `Surv(age1, age2)` scale, each term of the likelihood compares a death at age a to all those alive and under observation at age a .
- A Cox model on time-since-enrollment scale matches on time since enrollment. For the MCSA, this is the time since you got a letter in the mail.
- The splicing assumption ...

```
> plot(dsurv, cumhaz=2:4, fun= function(y) sqrt(y),
  col=1:2, lty=c(1,1,2,2,3,3),lwd=2, xmax=100,
  xlab="Age", ylab="Cumulative hazard")
```

3 Survival data: simple vs counting process vs timeline

The survival package started out with a simple `Surv(time, status)`, where status is 0/1 = censored/dead, or FALSE/TRUE, or 1/2.

When a need for time-dependent covariates arose we added the counting process form (`time1, time2, status, x1, x2, ...`):

- Over the interval $(\text{time1}, \text{time2}]$, this is the value of the covariates $x1, x2, \dots$
- `status` = an event at `time2`
- The square brackets are important, and reflect the fact that covariates need to be a predictable process.

$$PL = \prod_{i \in D} \frac{dN_i(t) e^{X_i(t)\beta}}{\sum_j Y_j(t) e^{X_j(t)\beta}} \quad (1)$$

$$= \prod_{i \in D} \frac{e^{X_i(t)\beta}}{\sum_j Y_j(t) e^{X_j(t)\beta}} dN_i(t) \quad (2)$$

A subject could have multiple rows, all but the last with status 0.

Then it was then pointed out that this form naturally accomodates time dependent strata. A couple years later we realized that it naturally handles the Andersen-Gill model. Later we realized we could do mult-state.

There are several oddities which will catch you.

1. The data for each subject needs to describe an actual observable timeline

2. Say a subject has observations as 0, 10, 36, 50 with diabetes at time 36. If diabetes is a covariate it will first be 1 in the (36,50] line, if it is an endpoint then in the (10,36] line of data: `tdc()` vs `event()` in `tmerge`
3. For each subject we first have to define a time window.
4. `Tmerge` calls have to be sequential

The `msm` and `mshmm` packages want a different form, which I call “timeline data”. Every row has an (id, time) pair, along with all the covariates known at that time. This data form is a bit odd for ordinary survival: everyone will have two lines, the first at time 0 with covariates and status= alive, the second at death or last fu, often with all covariates missing except status.

But for multistate data, the timeline data sets are much easier to create. (You don’t need `tmerge`). Much of the survival package can deal with either, I’m working on making this “all”.

```
> # make a counting process form of the dementia data
> # It is easiest to do it in "time since enroll" scale, then convert
> indx <- match(dem2$ptnum, dem2$ptnum)
> dem2$year <- dem2$age - dem2$age[indx]
>
> t1 <- subset(dem2, !duplicated(ptnum), c(ptnum, year, male, educ, any.e4))
> t2 <- subset(dem2, !duplicated(ptnum, fromLast=TRUE), c(ptnum, year,dx))
> dem2$dtemp <- with(dem2, nostutter(ptnum, dstate2, censor="alive"))
>
> dem2c <- tmerge(t1, t2, id=ptnum, death= event(year, dx=="death"))
> dem2c <- tmerge(dem2c, subset(dem2, duplicated(ptnum, fromLast=TRUE)),
  id=ptnum, cmc= tdc(year, cmc),
  centiloid= tdc(year, centiloid), p.tau181= tdc(year, p.tau181),
  hpvol= tdc(year, hpvol), memory= tdc(year, memory),
  global= tdc(year, global), dstate2 = event(year, dtemp),
  priordem= cumtdc(year, 1*(dstate=='dementia'))
> dem2c$dstate2[dem2c$death] <- "death"
> xx <- with(dem2, unlist(tapply(as.numeric(dstate), ptnum, cummax)))
> xx <- xx[duplicated(dem2$ptnum, fromLast=TRUE)]
> dem2c$cstate <- factor(xx,2:3, c("ND", "dementia"))
>
> subset(dem2, ptnum==14, c(ptnum, year, male, memory, dstate2))
  ptnum year male   memory dstate2
75    14  0.0   1 -3.258588      ND
76    14  1.2   1 -2.435467      ND
77    14  1.7   1      NA dementia
78    14  3.1   1      NA  death
> subset(dem2c, ptnum==14, c(ptnum, tstart, tstop, male, memory,
  cstate, dstate2, priordem))
  ptnum tstart tstop male   memory cstate dstate2 priordem
62    14    0.0   1.2   1 -3.258588     ND  alive      0
63    14    1.2   1.7   1 -2.435467     ND dementia    0
```

```

64    14    1.7    3.1    1 -2.435467 dementia    death    1
> indx <- match(dem2c$ptnum, dem2$ptnum)
> dem2c$age1 <- dem2c$ststart + dem2$age[indx]
> dem2c$age2 <- dem2c$ststop + dem2$age[indx]
>
> survcheck(Surv(age1, age2, dstate2) ~ 1, dem2c, id=ptnum, istate=cstate)
Call:
survcheck(formula = Surv(age1, age2, dstate2) ~ 1, data = dem2c,
  id = ptnum, istate = cstate)

```

Unique identifiers	Observations	Transitions
4317	21354	2123

Transitions table:

from	to		
	dementia	death	(censored)
ND	652	957	2708
dementia	0	514	138
death	0	0	0

Number of subjects with 0, 1, ... transitions to each state:

	count		
state	0	1	2
dementia	3665	652	0
death	2846	1471	0
(any)	2708	1095	514

Also interesting is the probability of ever having dementia. An easy trick to do this is to ignore any followup after a dementia diagnosis.

```

> dsurv3 <- survfit(Surv(age1, age2, dstate2) ~ male, dem2c, id= ptnum,
  istate= cstate,
  subset=(priordem==0), p0=c(1,0,0), start.time=65)
> plot(dsurv3[, -1], col=1:4, lwd=2, xmax=100,
  xlab="Age", ylab="Prob")
> legend(65, .5, c("Ever dem, F", "Ever dem, M", "Death w/o dem, F",
  "Death w/o dem, M"), col=1:4, lwd=2)

```

4 Assessing amyloid

A key question of the paper was the effect of amyloid levels on the progression to dementia. Is this expensive test worth doing? Key issues:

- Age is the most important covariate

- Presence of one or more $\epsilon 4$ alleles in the APOE gene are known to increase the risk of dementia. Some of that may be via increased amyloid deposition.
- There is a known sex*APOE interaction, with respect to the effect.
- About 40% of the subjects have amyloid. There was prior work using categorical amyloid.
- The PiB tracer did not become available until partway into the MCSA, those who have amyloid don't all start imaging at visit 0.

```
> temp <- ifelse(is.na(dem2c$centiloid), 0,
                 as.numeric(cut(dem2c$centiloid, c(-100, 22, 68, 400))))
> dem2c$agrp <- factor(temp, c(1:3, 0), c("low", "moderate", "high", "missing"))
> fit1 <- coxph(list(Surv(age1, age2, dstate2) ~ male + any.e4 + agrp + cmc,
                    1:2 ~ male*any.e4), dem2c, id= ptnum, istate=cstate)
```